Section A – Answer any 2 out of 3 questions in section A

Question 1

a)

(i)

The availability table of generators:

Availability (kW)	Probability (%)
230	83.79
180	1.71
150	9.31
130	4.41
Others	0.78

[1 mark]

Pn =150kW.

[1 mark]

$$\left(0.2+\frac{1-0.2}{\sqrt{n}}\right)n\times 10=150$$

[1 mark]

$\therefore n=47.4$
47 households

[1 mark]

(ii)

Option 1:
The availability table of generators:

Availability (kW)	Probability(%)
G1+G2+G3 ($\geq$ 150)	79.515
G2+G3 (180)	5.985
G1+G3 (=150)	8.835
Others	5.665

[2 marks]

$$P_1^C = 150-100 = 50\ \text{kW}$$

[1 marks]

$$c_1 = P_1^C \cdot 1000\ £/kw = 50000\ £$$

Option 2: Since the availability does not change, the capacity of G1 is the same as that in (i).

Model Answers

Some students did not notice the capacity of G1 should remain unchanged due to the same availability.

$c_2 = P_2^C \cdot 1500\ £/kw = 75000\ £$

[2 marks]

Option 1 is better.

[1 mark]

b)

Choose: $V_{B11} = 11kV,\ V_{B132} = 132kV, V_{B33} = 33kV$

[1 mark]

$P_r = 0.6p.u. \quad Q_r = 0.48p.u.$

[1 mark]

$X = X_1 + X_2 + X_3 = j0.15\frac{50}{60} + \frac{j100}{V_{B132}^2/S_B} + j0.12 = j0.532p.u.$

[1 mark]

$V_s = V_r + ji_rX = V_r + \frac{j(P_r - jQ_r)}{V_r^*}X$

$(V_s - V_r)V_r^* = j(P_r - jQ_r)X$

$V_s = \frac{15}{11}\angle 0° = 1.3636\angle 0° \qquad V_r = |V_r|\angle\theta$

$V_sV_r\cos\theta - |V_r|^2 = Q_rX; \qquad V_sV_r\sin\theta = -P_rX$

[2 marks]

Eliminate the angle: $|V_r|^4 + (2Q_rX - V_s^2)|V_r|^2 + (Q_r^2 + P_r^2)X^2 = 0$

$\therefore |V_{r1}| = 1.1003p.u. \qquad \therefore |V_{r2}| = 0.3715p.u.$

A few students eliminated the second solution without verification.

[1 mark]

$P_r = \frac{V_sV_r}{X}\sin(-\theta)$

$\theta_1 = -\sin^{-1}\frac{P_rX}{V_s|V_{r1}|} = -\sin^{-1}\frac{0.6\times0.532}{1.3636\times1.1003} = -12.28°$

$\theta_2 = -\sin^{-1}\frac{P_rX}{V_s|V_{r2}|} = -\sin^{-1}\frac{0.6\times0.532}{1.3636\times0.3715} = -39.06°$

[1 mark]

$i_1 = \frac{S_r^*}{V_{r1}^*} = \frac{0.6-j0.48}{1.1003\angle 12.28°} = 0.6983\angle -50.94°$

$i_2 = \frac{S_r^*}{V_{r2}^*} = \frac{0.6-j0.48}{0.3715\angle 39.06°} = 2.0683\angle -77.7198°$

[1 mark]

Model Answers

$S_{s1} = V_s i_1^* = 1.3636 \times 0.6983\angle 50.94° = 0.9522\angle 50.94° p.u.$

$|S_{s1}| = 47.6100MVA < 60MVA$

$S_{s2} = V_s i_2^* = 1.3636 \times 2.0683\angle 77.7198° = 2.8203\angle 77.7198° \, p.u.$

$|S_{s2}| = 141.0150MVA \gg 60MVA$, infeasible.

$\therefore V_r = 1.1003\angle -12.28° p.u. = 36.3099\angle -12.28° \, kV$

$P_s + jQ_s = 30MW + j36.9686MVar$

[2 marks]

Model Answers

Question 2

a)

(i)

$$S = V_r i_r^* = P_r + jQ_r = V_r \left(\frac{V_s - |V_r| \angle \theta}{R + jX} - |V_r| \angle \theta \frac{B}{2} \angle 90^\circ \right)^*$$

[1 mark]

$$= V_r \left(\frac{|V_s| - |V_r| cos\theta - j|V_r| sin\theta}{R + jX} - \frac{|V_r| B}{2} \angle (\theta + 90^\circ) \right)^*$$

$$= (|V_r| cos\theta + j|V_r| sin\theta) \left[\left(\frac{|V_s| R - |V_r| (R cos\theta + X sin\theta)}{R^2 + X^2} + \frac{|V_r| B}{2} sin\theta \right) \right.$$

$$\left. + j \left(\frac{|V_s| X + |V_r| (R \, sin\, \theta - X \, cos\theta)}{R^2 + X^2} + \frac{|V_r| B}{2} cos\theta \right) \right] = 0.4128 + j0.4203$$

[2 marks]

(ii)

$$S = V_r i_r^* = P_r + jQ_r = V_r \left(\frac{V_s - |V_r| \angle \theta}{R + jX} - |V_r| \angle \theta \frac{B}{2} \angle 90^\circ \right)^*$$

$$= V_r \left(\frac{|V_s| - |V_r| cos\theta - j|V_r| sin\theta}{R + jX} - \frac{|V_r| B}{2} \angle (\theta + 90^\circ) \right)^*$$

$$= (|V_r| cos\theta + j|V_r| sin\theta) \left[\left(\frac{|V_s| R - |V_r| (R cos\theta + X sin\theta)}{R^2 + X^2} + \frac{|V_r| B}{2} sin\theta \right) \right.$$

$$\left. + j \left(\frac{|V_s| X + |V_r| (R \, sin\, \theta - X \, cos\theta)}{R^2 + X^2} + \frac{|V_r| B}{2} cos\theta \right) \right] = 0.3869 + j0.7718$$

[1 mark]

R<<X, difference in voltage magnitude is mainly related to reactive power flow

[1 mark]

(iii)

$$S = V_r i_r^* = P_r + jQ_r = V_r \left(\frac{V_s - |V_r| \angle \theta}{R + jX} - |V_r| \angle \theta \frac{B}{2} \angle 90^\circ \right)^*$$

Model Answers

$$= V_r\left(\frac{|V_s| - |V_r|cos\theta - j|V_r|sin\theta}{R + jX} - \frac{|V_r|B}{2}\angle(\theta + 90^\circ)\right)^*$$

$$= (|V_r|cos\theta + j|V_r|sin\theta)\left[\left(\frac{|V_s|R - |V_r|(Rcos\theta + Xsin\theta)}{R^2 + X^2} + \frac{|V_r|B}{2}sin\theta\right)\right.$$

$$\left. + j\left(\frac{|V_s|X + |V_r|(R\,sin\,\theta - X\,cos\theta)}{R^2 + X^2} + \frac{|V_r|B}{2}cos\theta\right)\right] = 0.7985 + j0.3498$$

[1 mark]

R<<X, difference in voltage angle is mainly related to active power flow

[1 mark]

b)

Method 1:

$$E_{ab} = I_A Z_L + I_{AB} Z_\Delta - I_B Z_L = (I_A - I_B)Z_L + I_{AB} Z_\Delta$$
$$I_A - I_B = (I_{AB} - I_{CA}) - (I_{BC} - I_{AB}) = 2I_{AB} - I_{BC} - I_{CA}$$
$$I_{AB} + I_{BC} + I_{CA} = 0$$

[3 marks]

Combine the above 3 equations:

$$E_{ab} = 3I_{AB} Z_L + I_{AB} Z_\Delta$$

[1 mark]

Load currents:

$$I_{AB} = \frac{E_{ab}}{3Z_L + Z_\Delta} = 8.5129\angle - 33.96°A$$
$$I_{BC} = 8.5129\angle - 153.96°A$$
$$I_{AB} = 8.5129\angle 86.04°A$$

[3 marks]

Line currents:

$$I_A = I_{AB} - I_{CA} = 14.7448\angle - 63.96°A$$
$$I_B = 14.7448\angle 176.04°A$$
$$I_C = 14.7448\angle 56.04°A$$

[3 marks]

Voltages at load terminals:

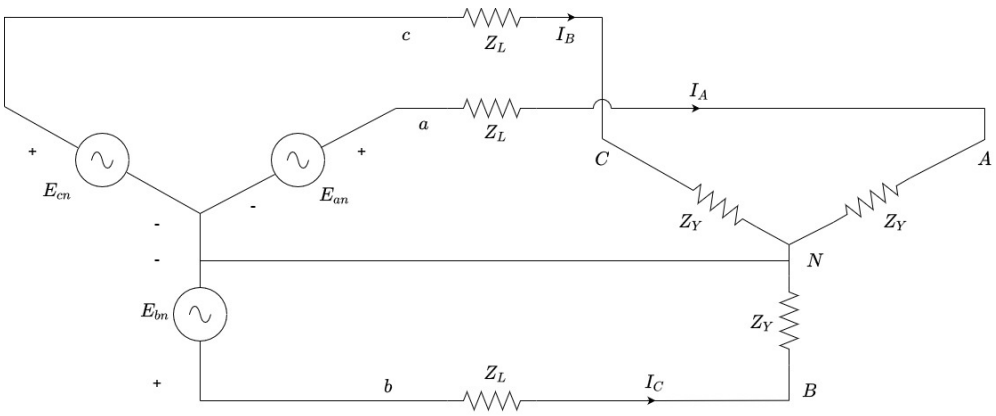
$$E_{AB} = Z_\Delta I_{AB} = 36\angle 30°(8.5129\angle - 33.96°) = 306.4644\angle - 3.96°V$$
$$E_{BC} = 306.4644\angle - 123.96°$$
$$E_{CA} = 306.4644\angle 116.04°$$

[3 marks]

Method 2:

Convert the Δ-load to an equivalent Y load:

Model Answers



$$Z_Y = 12\angle 30°\ \Omega$$

[4 marks]

Line currents:

$$I_A = \frac{E_{an}}{Z_L + Z_Y} = 14.7448\angle -63.96°A$$

$$I_B = 14.7448\angle 176.04°A$$

$$I_C = 14.7448\angle 56.04°A$$

[3 marks]

The Δ-load currents:

$$I_{AB} = \frac{I_A}{\sqrt{3}}\angle 30° = 8.5129\angle -33.96°A$$

$$I_{BC} = 8.5129\angle -153.96°A$$

$$I_{AB} = 8.5129\angle 86.04°A$$

[3marks]

Voltages at load terminals:

$$E_{AB} = Z_\Delta I_{AB} = 36\angle 30°(8.5129\angle -33.96°) = 306.4644\angle -3.96°V$$

$$E_{BC} = 306.4644\angle -123.96°$$

$$E_{CA} = 306.4644\angle 116.04°$$

[3 marks]

Model Answers

Question 3

a)

$$i_1 = \frac{1}{Z_1}(V_1 - V_2) = \frac{1}{Z_1}V_1 - \frac{1}{Z_1}V_2$$

$$i_2 = \frac{1}{Z_1}(V_2 - V_1) + \frac{1}{Z_2}(V_2 - V_3) + \frac{1}{Z_4}(V_2 - V_4) + \frac{1}{Z_5}(V_2 - V_4)$$

$$= -\frac{1}{Z_1}V_1 + \left(\frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_4} + \frac{1}{Z_5}\right)V_2 - \frac{1}{Z_2}V_3 - (\frac{1}{Z_4} + \frac{1}{Z_5})V_4$$

$$i_3 = \frac{1}{Z_2}(V_3 - V_2) + \frac{1}{Z_3}(V_3 - V_4) = -\frac{1}{Z_2}V_2 + \left(\frac{1}{Z_2} + \frac{1}{Z_3}\right)V_3 - \frac{1}{Z_3}V_4$$

$$i_4 = \frac{1}{Z_4}(V_4 - V_2) + \frac{1}{Z_5}(V_4 - V_2) + \frac{1}{Z_3}(V_4 - V_3) = -\left(\frac{1}{Z_4} + \frac{1}{Z_5}\right)V_2 - \frac{1}{Z_3}V_3 + \left(\frac{1}{Z_3} + \frac{1}{Z_4} + \frac{1}{Z_5}\right)V_4$$

[4 marks]

$$\therefore Y = \begin{pmatrix} \frac{1}{Z_1} & -\frac{1}{Z_1} & 0 & 0 \\ -\frac{1}{Z_1} & \frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_4} + \frac{1}{Z_5} & -\frac{1}{Z_2} & -(\frac{1}{Z_4} + \frac{1}{Z_5}) \\ 0 & -\frac{1}{Z_2} & \frac{1}{Z_2} + \frac{1}{Z_3} & -\frac{1}{Z_3} \\ 0 & -\left(\frac{1}{Z_4} + \frac{1}{Z_5}\right) & -\frac{1}{Z_3} & \frac{1}{Z_3} + \frac{1}{Z_4} + \frac{1}{Z_5} \end{pmatrix}$$

[2 marks]

Some students did not express Y matrix in terms of Zi.

b)

$$V_1 = 1\angle 0° \qquad V_2^0 = 1\angle 0° \qquad V_3^0 = 1.1\angle 0° \qquad V_4^0 = 1\angle 0°$$

[1 mark]

$$Y = \begin{pmatrix} -j4 & j4 & 0 & 0 \\ j4 & -j13 & j4 & j5 \\ 0 & j4 & -j9 & j5 \\ 0 & j5 & j5 & -j10 \end{pmatrix}$$

[1 mark]

Iteration 1:

Some students made mistakes in complex number calculation.

$$i_2 = \frac{S_2^*}{V_2^*} = Y_{21}V_1 + Y_{22}V_2 + Y_{23}V_3 + Y_{24}V_4$$

$$V_2^1 = \frac{1}{Y_{22}}\left(\frac{S_2^*}{V_2^{0*}} - Y_{21}V_1 - Y_{23}V_3^0 - Y_{24}V_4^0\right) = \frac{1}{-j13}\left(\frac{-0.95 + j}{1} - j4 - j4 \times 1.1 \ - j5\right)$$

$$= 0.9566\angle - 4.3810°$$

[2 marks]

$$S_3^* = V_3^{0*}i_3 = V_3^{0*}(Y_{31}V_1 + Y_{32}V_2^1 + Y_{33}V_3^0 + Y_{34}V_4^0)$$

[1 mark]

$$S_3^{1*} = 1.05 - j1.1931$$

Model Answers

[1 mark]

$$V_3^1 = \frac{1}{Y_{33}}\left(\frac{S_3^{1*}}{V_3^{0*}} - Y_{31}V_1 - Y_{32}V_2^1 - Y_{34}V_4^0\right)$$
$$= \frac{1}{-j9}\left(\frac{1.05 - j1.1931}{1.1} - 0 - j4(0.9566\angle -4.3810°) - j5\right) = 1.1025\angle 3.8270°$$

[1 mark]

$$V_3^1 = 1.1\angle 3.8270°$$

[1 mark]

$$V_4^1 = \frac{1}{Y_{44}}\left(\frac{S_4^*}{V_4^{0*}} - Y_{41}V_1 - Y_{42}V_2^1 - Y_{43}V_3^1\right)$$
$$= \frac{1}{-j10}\left(\frac{-1.05 + j}{1} - 0 - j5(0.9566\angle -4.3810°) - j5(1.1\angle 3.8270°)\right)$$
$$= 0.9328\angle -6.4474°$$

[2 marks]

c)

To reduce the power flow between 1 and 2, generator in Bus 1 needs to reduce the output and at the same time generator in Bus 3 needs to increase the output.

[2 marks]

As the decrease of output in Bus 1, phase angle and/or the amplitude of the voltage in Bus 2 will decrease.

[2 marks]

Most of the students failed to answer part c) correctly.

Model Answers

Question 4

a)

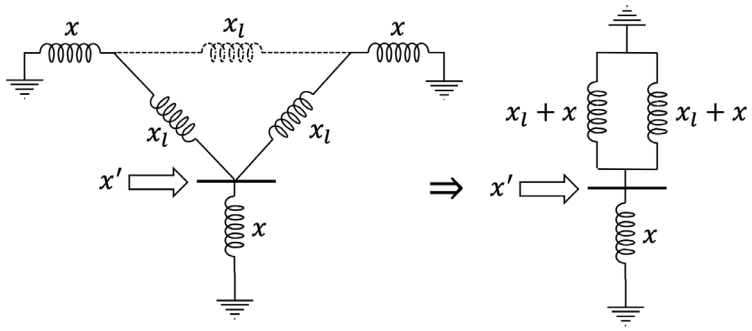
$S_{base} = 100$ MVA

Equivalent reactance at each bus

$$x = \frac{S_{base}}{SCL} = \frac{100}{2500} = 0.04 \text{ p.u.}$$

[1 mark]

Equivalent circuit looking into each bus



[3 marks]

A few students did not show the equivalent circuit and subsequent simplification explicitly.

Due to the symmetry in the circuit, one transmission line (shown by dotted) won't play a part in the equivalent reactance

The students didn't spot the circuit symmetry and went for star-delta conversion which wasn't necessary.

[1 mark]

Equivalent reactance after interconnection

$$x' = x \parallel [(x_l + x) \parallel (x_l + x)]$$
$$= \frac{x(x + x_l)}{(3x + x_l)} = 0.0255 \text{ p.u.}$$

[2 marks]

Short-circuit levels after interconnection

$$SCL'_1 = SCL'_2 = SCL'_3 = \frac{S_{base}}{x'} = 3928.6 \text{ MVA}$$

[1 mark]

Model Answers

A few students used bus admittance matrix to answer Q4(a) while it was clearly stated that this part was supposed to be answered using circuit simplification.

b)

Equivalent reactance at bus 3

$$x_3' = x \parallel [(x_l + x) \parallel (x_l + x)]$$
$$= \frac{x(x + x_l)}{(3x + x_l)} = 0.0255 \text{ p.u.}$$

[1 mark]

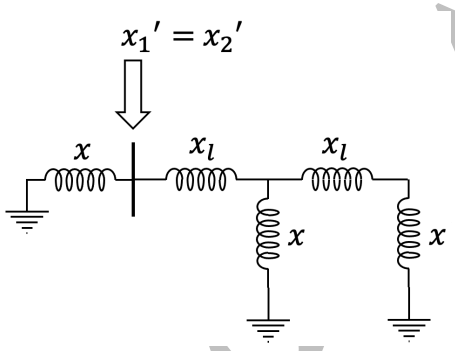
Short-circuit level at bus 3

$$SCL_3' = \frac{S_{base}}{x_3'} = 3928.6 \text{ MVA}$$

Remains unchanged from part (i) as the reactance of line connecting systems S_1 and S_2 does not play a part in the equivalent reactance of S_3

[1 mark]

Equivalent reactance at bus 1 and 2



[1 mark]

$$x_1' = x_2' = x \parallel [x_l + \{x \parallel (x_l + x)\}] = \frac{x({x_l}^2 + x^2 + 3xx_l)}{3x^2 + {x_l}^2 + 4xx_l} = 0.0306 \text{ p.u.}$$

[2 marks]

Short-circuit level at bus 1 and 2

$$SCL_1' = SCL_2' = \frac{S_{base}}{x_1'} = 3262.7 \text{ MVA}$$

Short circuit level at bus 1 and 2 reduces when the interconnection between $S_1$and S_2 is out of operation.

[1 mark]

Model Answers

c)

Line admittance $\frac{1}{x_l} = -j10$ p.u.

System equivalent admittance $\frac{1}{x} = -j25$ p.u.

[1 mark]

Bus admittance matrix is given by

$$Y_b = \begin{bmatrix} -j10 - j10 - j25 & j10 & j10 \\ j10 & -j10 - j10 - j25 & j10 \\ j10 & j10 & -j10 - j10 - j25 \end{bmatrix} \text{p.u.}$$

[2 marks]

Some students didn't add the system equivalent admittance in the diagonal terms.

Bus impedance matrix

$$Z_b = {Y_b}^{-1} = \begin{bmatrix} j0.0255 & j0073 & j0073 \\ j0073 & j0.0255 & j0073 \\ j0073 & j0073 & j0.0255 \end{bmatrix} \text{p.u.}$$

[2 marks]

Equivalent reactance at each bus is given by the diagonal elements of the bus impedance matrix.

$x' = Z_b(1,1) = Z_b(2,2) = Z_b(3,3) = j0.0255$ p.u.

Same as the result obtained in part (i)

[1 mark]

Model Answers

Question 5

a)

i) Line to ground fault

Equivalent sequence impedance at fault point

$Z_{1eq} = jX_1 \parallel jX_1 = 0.5 \times jX_1 = j0.125$ p.u.

$Z_{2eq} = jX_2 \parallel jX_2 = 0.5 \times jX_2 = j0.085$ p.u.

[1 mark]

The generator with floating neutral won't feature in the zero-sequence network

$Z_{0eq} = 3R_n + jX_0 = 0.9 + j0.05$ p.u.

[2 marks]

Some students either mistakenly added the neutral grounding resistance as reactance or just forgot to add it all together. This error percolated into the following calculations and also in part (iii).

Fault current for LG fault

$$I_f = \frac{3}{Z_{1eq} + Z_{2eq} + Z_{0eq}} = 3.07 - j0.89 \text{ p.u.} = 3.2\angle -16.11° \text{ p.u.}$$

[1 mark]

Base current

$$I_b = \frac{S_b}{\sqrt{3}V_b} = \frac{25}{\sqrt{3} \times 13} = 1.11 \text{ kA}$$

Fault current magnitude

$|I_f| \times I_b = 3.56$ kA

[1 mark]

ii) Line to line fault

Positive sequence current

$$I_{a1} = \frac{1}{Z_{1eq} + Z_{2eq}} = -j4.76 \text{ p.u.}$$

[1 mark]

Fault current

$I_f = -j\sqrt{3}I_{a1} = -8.25$ p.u.

Model Answers

[1 mark]
A few students forgot the calculate the fault current in phase domain.

Fault current magnitude
$|I_f| \times I_b = 9.16$ kA

[0.5 mark]

$V_{a1} = 1 - I_{a1}Z_{1eq} = 0.4$ p.u. $= V_{a2}$

[1 mark]

Voltage at healthy phase

$V_a = V_{a1} + V_{a2} + V_{a0} = 2V_{a1} = 0.81$ p.u.

[1 mark]

Magnitude of the healthy phase voltage

$|V_a| \times V_{base} = 10.52$ kV

[0.5 mark]

iii) Line to line to ground fault

Positive sequence current

$$I_{a1} = \frac{1}{Z_{1eq} + Z_{2eq} \parallel Z_{0eq}} = 0.18 - j4.78 \text{ p.u.}$$

[1 mark]

$V_{a1} = 1 - I_{a1}Z_{1eq} = 0.4 - j0.02$ p.u.

[1 mark]

$$I_{a0} = -\frac{V_{a1}}{Z_{0eq}} = -0.44 + j0.05 \text{ p.u.}$$

Fault current

$I_f = 3I_{a0} = -1.33 + j0.15$ p.u.$=1.34\angle 173.62°$ p.u.

Fault current magnitude
$|I_f| \times I_b = 1.49$ kA

Model Answers

[2 marks]

$V_a = V_{a1} + V_{a2} + V_{a0} = 3V_{a1} = 1.21 - j0.07$ p.u.=1.21∠-3.2° p.u.

Magnitude of the healthy phase voltage

$|V_a| \times V_{base} = 15.71$ kV

[1 mark]

b)

Fault condition in phase domain

$I_a = 0$

$V_{bb'} = V_{cc'} = 0$

[2 marks]

In sequence domain

$I_{a1} + I_{a2} + I_{a0} = 0$

$V_{aa'1} = V_{bb'1} = V_{cc'1} = \frac{1}{3}V_{aa'}$

[2 marks]

The equations in sequence domain reflect a parallel connection between the three sequence networks

[1 mark]

A number of students failed to answer this part (b) correctly.

Model Answers

Question 6

a)

$$\frac{P'_{max}}{P_{max}} = \frac{X}{X(1-k)} = 2$$

$k = 0.5$ or 50%

[2 marks]

b)

$\delta_{max} = 90°$

$$Q_C = \frac{2V^2 k}{X(1-k)^2}(1 - \cos\delta_{max}) = 20.0 \text{ p.u.}$$

[2 marks]

c)

$$P_{max0} = \frac{1}{X} = 5.0 \text{ p.u.}$$

$P = 3$ p.u.

$$\delta_0 = \sin^{-1}\left(\frac{P}{P_{max0}}\right) = 36.87°$$

$$\delta_{cr} = \cos^{-1}[(\pi - 2\delta_0)\sin\delta_0 - \cos\delta_0] = 71.77°$$

[1 mark for each step – 4 marks]

d)

$k = 0.3$

$$P_{max1} = \frac{1}{X(1-k)} = 7.14 \text{ p.u.}$$

[1 mark]

$$\delta_0 = \sin^{-1}\left(\frac{P}{P_{max1}}\right) = 24.83°$$

[1 mark]

$$\delta_{cr} = \cos^{-1}[(\pi - 2\delta_0)\sin\delta_0 - \cos\delta_0] = 87.26°$$

[1 mark]

Model Answers

e)

For the situation in part (c)

$$t_{cr1} = \sqrt{\frac{4H(\delta_{cr} - \delta_0)}{2\pi f P}} = 113.7 \text{ ms}$$

[2 marks]

For the situation in part (d)

$$t_{cr1} = \sqrt{\frac{4H(\delta_{cr} - \delta_0)}{2\pi f P}} = 152.1 \text{ ms}$$

The critical clearing time increases by $(152.1 - 113.7)$ ms $= 38.4$ ms

[2 marks]

f)

For the situation in part (c)

$$\frac{\partial P}{\partial \delta} = P_{max0} \cos \delta_0 = 4 \text{ p.u.}$$

$$f_n = \frac{1}{2\pi} \sqrt{\frac{2\pi f}{2H} \left(\frac{\partial P}{\partial \delta}\right)} = 1.78 \text{ Hz}$$

[1 mark]

For the situation in part (d)

$$\frac{\partial P}{\partial \delta} = P_{max1} \cos \delta_0 = 6.48 \text{ p.u.}$$

$$f_n = \frac{1}{2\pi} \sqrt{\frac{2\pi f}{2H} \left(\frac{\partial P}{\partial \delta}\right)} = 2.27 \text{ Hz}$$

[1 mark]

g)

Under the condition specified in the problem, the critical clearing angle (δ_{cr}) depends only on the initial power angle (δ_0) which depend on $\frac{P}{P_{max}}$

[1 mark]

In part (c)

$$\frac{P}{P_{max}} = \frac{3}{5} = 0.6$$

[1 mark]

Model Answers

If P' is the power transfer possible with the same critical clearing angle (δ_{cr}) as in part (c)

$$\frac{P'}{P_{max1}} = 0.6 \rightarrow P' = 0.6 \times P_{max1} = 4.29 \text{ p.u.}$$

[1 mark]

Most students approached Q6 correctly but a few made spurious mistakes in fitting numbers to the expressions.

Model Answers